University of Colombo, Sri Lanka

University of Colombo School of Computing

DEGREE OF BACHELOR OF INFORMATION TECHNOLOGY (EXTERNAL)

Academic Year 2023— 2nd Year Examination — Semester 3

IT3206 — Data Structures and Algorithms

Part 1 - Multiple Choice Question Paper

(2 Hours for both Part 1 and Part 2)

Important Instructions

- This paper has **two (2) parts, Part 1 and Part 2**.
- The total duration of **both Part 1 and Part 2 is 2 hours**.
- The medium of instructions and questions is English.
- This paper (Part 1) has **25 MCQ questions** on **8 pages**. Answer **all** questions.
- Each question will have **5 (five)** choices with **ONE OR MORE** correct answers.
- There will be a penalty for incorrect responses to discourage guessing.
- The mark given for a question will vary from -1 (All the incorrect choices are marked & no correct choices are marked) to +1 (All the correct choices are marked & no incorrect choices are marked). However, **the minimum mark per question would be zero**.
- Answers should be marked on the **special answer sheet** provided.
- Note that questions appear on both sides of the paper. If a page or part of a page is not printed, please inform the supervisor/invigilator immediately.
- Mark the correct choices on the question paper first and then transfer them to the given answer sheet which will be machine marked. **Please completely read and follow the instructions given on the other side of the answer sheet before you shade your correct choices.**
- Any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, is not allowed.
- Calculators are **not** allowed.
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1). Which of the following statement(s) is/are correct regarding data structures?

a). Arrays are heterogeneous data structures.
b). Each data element type in a homogenous data structure is the same.
c). Data elements are attached hierarchically in a linear data structure.
d). Dynamic data structures consist of fixed, permanent structures and sizes at compile time.
e). Static data structures have fixed formats and sizes.

2). When implementing a circular queue with an array of size MAX_SIZE, it becomes full when

a). front = (rear + 1)
b). front = rear - 1
c). rear = front+1
d). front = 0 and rear = MAX_SIZE - 1
e). rear = MAX_SIZE + 1

3). Which of the following statement(s) is/are correct on skip lists?

I. Non-sequential searching can be done with a skip list.
II. Skip list is a linear structure.
III. Skip lists can be used to implement priority queues that work very well in multi-threaded environments.
IV. An array is an efficient data structure to implement a skip list.

a). (I) only. b). (I) and (II) only. c). (I) and (III) only.
d). (I) and (IV) only. e). (I), (III), and (IV) only.

4). Consider the following pseudo code:

```
items = Stack ()
for (int i=0;i<19;i++){
 if (i % 3 == 0 or i % 4==0 )
     items.push( i )
 elseif (i % 5 == 0 )
     items.pop()
}
```

When the execution is finished, what would be the final content in the stack?

a). 3,6,16,18 b). 0,3,6,8,18 c). 0,3,6,12,16,18
d). 0,3,4,6,12,16,18 e). 0,3,6,8,12,15,16,18

5). The number of elements in the adjacency matrix of a graph having n vertices is

a). n^2 b). n log (n) c). log (n)
d). 2n-1 e). n+1

6). Which of the following statement(s) is/are correct regarding sorting algorithms?

a). Insertion Sort makes no more than O(n log n) pair-wise comparisons.
b). Merge sort is not an in-place sorting algorithm.
c). Selection sort is not a default stable sorting algorithm.
d). The best-case runtime complexity of merge sort is O(n).
e). The worst-case runtime complexity of bubble sort is $O(n^2)$.

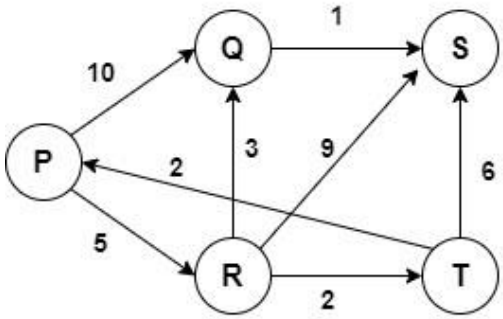
7). Consider the following function `func1()`

```
public void func1(){
      Node pointer = head;
      Node previous = null, current = null;
      while (pointer != null){
         current = pointer;
         pointer = pointer.next;
         current.next = previous;
         previous = current;
         head = current;
      }
   }
```

What is the above function intended to do?

a). Deleting an element from a linked list.
b). Inserting a new element into a linked list.
c). Reversing the elements in the linked list.
d). Searching for an element in a linked list.
e). Sorting elements in the linked list.

8). What is the shortest path from the **P** node to the **S** node?

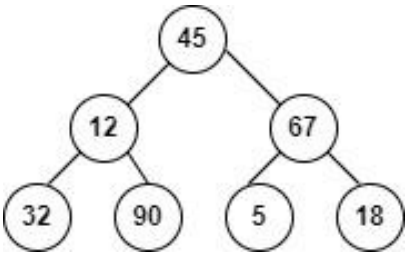


a). P->R->S
b). P->Q->R->S
c). P->R->Q->S
d). P->R->T->S
e). P->T->R->S

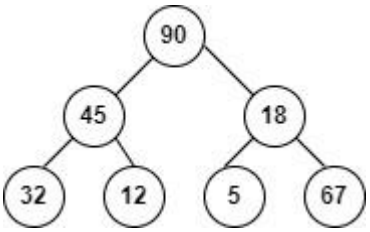
9). Which of the following sorting algorithms uses the divide-and-conquer approach to sorting?

a). Bubble Sort
b). Insertion Sort
c). Merge Sort
d). Quick Sort
e). Selection Sort

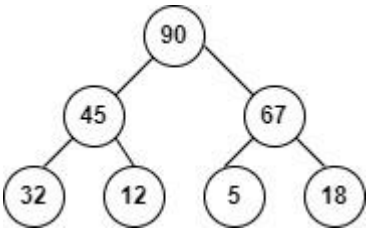
10). Consider the following Binary Tree:



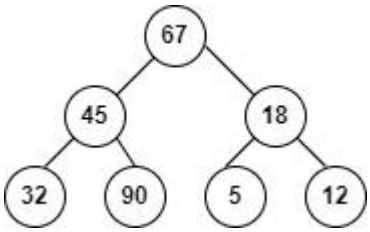
After building a max heap, what will be the correct max heap?



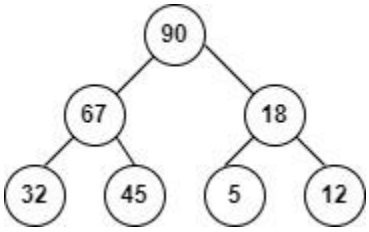
a).



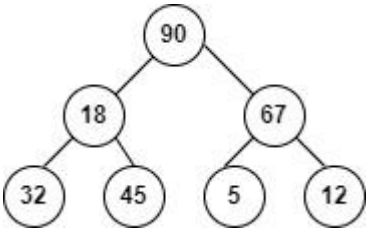
b).



c).



d).



e).

11). When the program below executes, what results would it produce?

```
public static void main(String[] args) {
     char [] x = {'a', 'b', 'c', 'd', 'e'};
     func(x, 5);
}
public static void func(char[] x, int length) {
     System.out.print(" " + x[length - 1]);
     func(x, length - 1);
}
```

a). The program displays a b c d e.
b). The program displays e d c b a.
c). The program displays e e e e e.
d). The program displays a b c d e, then raises an `ArrayIndexOutOfBoundsException`.
e). The program displays e d c b a, then raises an `ArrayIndexOutOfBoundsException`.

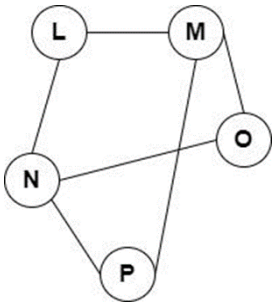
12). The following elements need to be sorted in ascending order.

[22, 62, 52, 39, 18, 94, 72, 37]

How many passes will be required if the Bubble Sort technique is used?

a). 3	b). 4	c). 5
d). 6	e). 7	

13). Which of the following traversals is/are true regarding the Breadth First Traversal (BFT) applied Graph (G) starting at node L?



a). L M O P N
b). L M N O P
c). L N M O P
d). L N M P O
e). L N P O M

14). "Removing a node in a linked list means removing the node from that address and not removing the contents of the node." Select correct statement(s) from below. Assume both singly and doubly linked lists are not empty after deleting a node.

I. A node in a singly linked list can be deleted in O(n) time if a pointer/link to the previous node is provided.
II. A node in a singly linked list can be deleted in O(1) time if the pointer/link to the node is provided.
III. A node in a singly linked list can be deleted in O(1) time if a pointer/link to the previous node is provided.
IV. A node in a doubly linked list can be deleted in O(1) time if a pointer/link to the node is provided.

a). (I) only.	b). (I) and (II) only.	c). (I) and (III) only.
d). (III) and (iv) only.	e). (I),(III) and (iv) only.	

15). What would be returned from the function below if it was called with the value n = 10?

```
int Func(int n){
     if (n == 0)
          return 1;
     else
          return (n * Func(n-2));
}
```

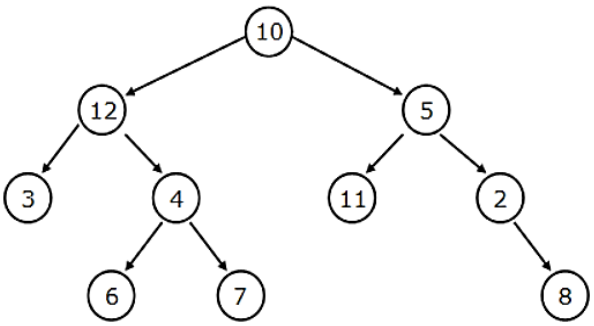
a). 3620	b). 3840	c). 38480
d). 362880	e). 382800	

16). What would be the output of the below code?

```
Queue q = new Queue();
q.enqueue(0);
q.enqueue(1);
for (int i = 0; i < 9; i++) {
    int x = dequeue();
    System.out.println(x);
    int y = q.front();
    q.enqueue(x + y);
}
```

a). 1, 2, 3, 5, 8, 13, 21.
b). 1, 2, 3, 5, 8, 13, 21, 34.
c). 1, 1, 2, 3, 5, 8, 13, 21.
d). 0, 1, 1, 2, 3, 5, 8, 13, 21.
e). 0, 1, 1, 2, 3, 5, 8, 13, 21, 34.

17). For the below tree, which are the appropriate **post-order** and **in-order** traversals?



a). (3,6,7,4,12,11,8,2,5,10) and (10,12,3,4,6,7,5,11,2,8) respectively.
b). (3,6,7,4,12,11,8,2,5,10) and (3,12,6,4,7,10,11,5,2,8) respectively.
c). (3,12,6,4,7,10,11,5,2,8) and (3,6,7,4,12,11,8,2,5,10) respectively.
d). (3,12,6,4,7,10,11,5,2,8) and (10,12,3,4,6,7,5,11,2,8) respectively.
e). (10,12,3,4,6,7,5,11,2,8) and (3,12,6,4,7,10,11,5,2,8) respectively.

18). Which of the following statement(s) is/are correct regarding a tree data structure?

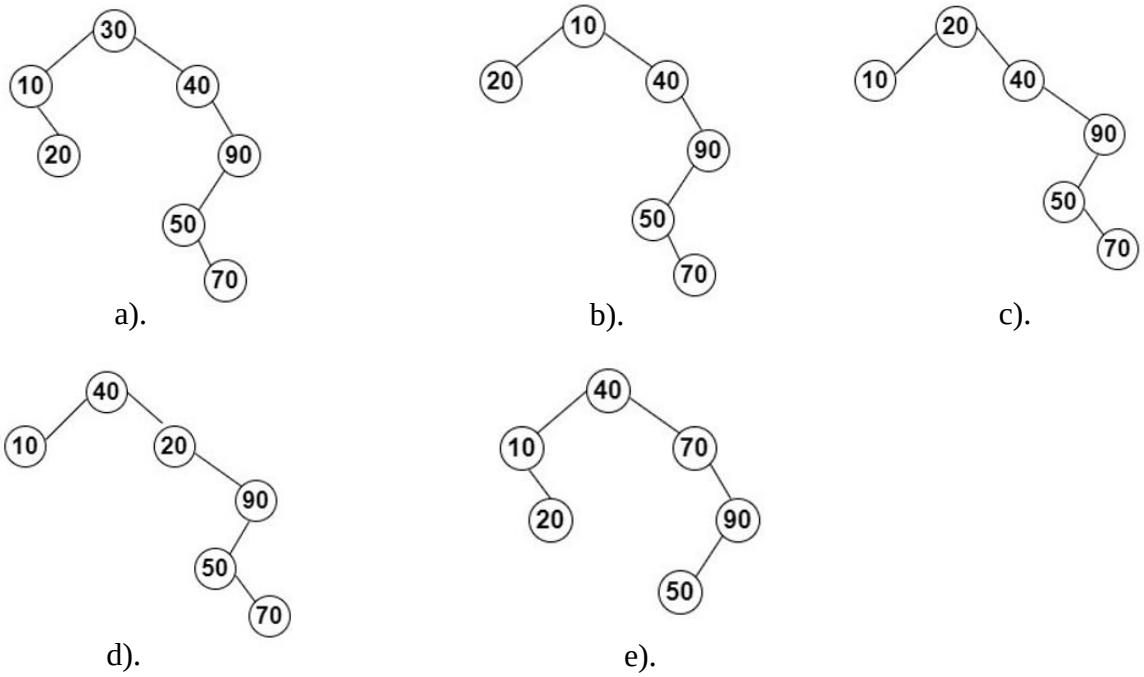
a). A binary heap is a complete binary tree.
b). A full binary tree is a binary tree in which each node has exactly two children.
c). In an AVL tree, any node's height difference between the left and right subtrees is limited to 0, 1, and 2.
d). In a balanced Binary Search Tree, the time complexity for searching an element is O(log(n)).
e). Trees can be used to evaluate arithmetic operations.

Answer questions **19** and **20** by considering a binary search tree with the following elements added to it in the prescribed order: 30, 10, 40, 90, 20, 50, and 70.

19). What is the leftmost node of the right subtree?

a). 20	b). 70	c). 50
d). 10	e). 90	

20). What would be the resulting tree after deleting the root node?



21). Which of the following is/are **not** an application(s) of circular linked lists?

a). Arithmetic expression evaluation.	b). Hash table implementation.	c). Managing function calls.
d). Playlists in music and media players.	e). Task scheduling.	

22). What is the running time complexity of the following code?

```
int i, j, k = 0;
for (i = n / 2; i <= n; i++)
{
      for (j = 2; j <= n; j = j * 2)
      {
          k = k + n / 2;
      }
}
```

a). O(n)	b). $O(n^2)$	c). O(log n)
d). O(n log n)	e). $O(n^2 \log n)$	

23). In a connected undirected Graph G, if you perform a depth-first search (DFS) starting from any vertex, what can be assured?

a). DFS may not visit all vertices.
b). DFS may result in an infinite loop.
c). DFS will always visit all vertices at least twice.
d). DFS will visit all vertices and edges exactly once.
e). DFS will visit all vertices and edges, but not necessarily exactly once.

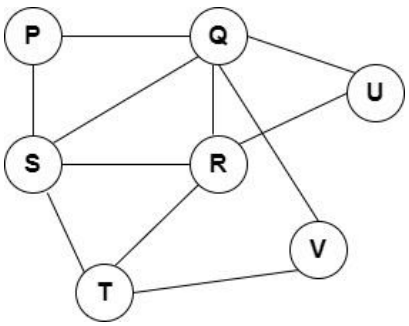
24). Consider the following pseudo code:

```
function EVALUATE(expression)
    S = empty stack
    for each token in the expression do
        if the token is a number
            push(S, token)
        else
            operand1 = pop(S)
            operand2 = pop(S)
            push(S, operand2 (token) operand1)
         end if
      end for
      return top of S
```

If the expression is provided as **281*+4-**, what would be the result?

a). 0 b). 6 c). 12
d). 14 e). 26

25). Consider the following graph:



If we run a breadth-first search (BFS) on above graph starting at any vertex, which one of the following is a possible order for visiting the nodes?

a). P, R, Q, T, S, V,U
b). T, S, R, V, P, Q, U
c). U, Q, T, P, V, R, S
d). V, T, P, Q, S, R, U
e). V, T, U, R, S, P, Q

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